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data/
├─ raw/                                as-downloaded archives (read-only)
├─ validated/
│   ├─ BTCUSDT/5m.parquet              683,424 rows (listing → cutoff)
│   ├─ BTCUSDT/fundingRate.parquet    7,119 events
│   └─ ETHUSDT/...                    same layout per symbol
├─ processed/
│   ├─ BTCUSDT/1h_development.parquet 52,608 bars (2020-01 → 2025-12)
│   ├─ BTCUSDT/1h_holdout.parquet     4,344 bars (frozen)
│   └─ .../{15m,1h}_{development,holdout}.parquet
├─ manifests/
│   └─ binance_um_<sym>_<stream>[_tf][_part].json
│       {row_count, period, data_sha256, git_commit, code_version}
└─ catalog/catalog.sqlite             experiment registry (Fig. 4.8)

```

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artifacts/runs/<kind>_<UTC-stamp>_<6-hex>/
├─ resolved_experiment_config.yaml · search_space.json · folds.json
├─ seed_schedule.json · git_state.json · environment.json
├─ {random_search,genetic_algorithm}_candidates.parquet
├─ ..._fold{0..14}_test_{equity,trades}.parquet
└─ comparison_summary.json · dataset_manifests.json

```

Integrity model: paths are deterministic and human-readable; integrity is enforced by SHA-256 recorded in per-dataset manifests and re-verified against the bytes — not by content-addressed paths. A mismatch fails the run loudly.